$\bar{a}'$ as a state variable with the motion control law of TIE 2025

2026-11-06 z axis

1 Motion model and control law

Normalized variables

$$\bar{x} = \frac{x}{R}$$
$$\bar{a} = \frac{a_x}{a_{\text{nom}}}$$
$$\bar{f} = a_o\, f$$
$$a_{\text{nom}} = \frac{\Delta t}{\gamma_N}$$
$$a_o = \frac{a_{\text{nom}}}{R}$$

Motion model

$$\bar{x}[k+1] = \bar{x}[k] + \bar{a}[k]\left(\bar{f}_d[k] + \bar{f}_T[k]\right)$$

Tracking error, control objective, measured tracking error

$$\delta\bar{x}[k] = \bar{x}_d[k] - \bar{x}[k]$$
$$\delta\bar{x}[k+1] = \lambda_c\, \delta\bar{x}[k]$$
$$\bar{x}_m[k] = \bar{x}[k-d] + n_{\bar{x}}[k]$$
$$\delta\bar{x}_m[k] = \bar{x}_d[k-d] - \bar{x}_m[k] \qquad (d=2)$$

Motion control law

$$\Delta\bar{x}_d[k] = \bar{x}_d[k+1] - \bar{x}_d[k]$$
$$\bar{f}_d[k] = \hat{\bar{a}}^{-1}[k]\left\{\Delta\bar{x}_d[k] + (1-\lambda_c)\,\delta\hat{\bar{x}}_3[k]\right\}$$
$$\hat{M}[k] = \Delta\bar{x}_d[k] + (1-\lambda_c)\,\delta\hat{\bar{x}}_3[k]$$
$$\hat{M}[k] = \hat{\bar{a}}[k]\,\bar{f}_d[k]$$

2 Dynamics of tracking error

$$e_{\bar{a}}[k] = \bar{a}[k] - \hat{\bar{a}}[k]$$
$$e_{\delta\bar{x}_3}[k] = \delta\bar{x}[k] - \delta\hat{\bar{x}}_3[k]$$

$$\delta\bar{x}[k+1] = \delta\bar{x}[k] + \Delta\bar{x}_d[k] - \bar{a}[k]\,\bar{f}_d[k] - \bar{a}[k]\,\bar{f}_T[k]$$
$$\bar{a}[k]\,\bar{f}_d[k] = \hat{\bar{a}}[k]\,\bar{f}_d[k] + e_{\bar{a}}[k]\,\bar{f}_d[k]$$
$$\bar{a}[k]\,\bar{f}_d[k] = \Delta\bar{x}_d[k] + (1-\lambda_c)\,\delta\hat{\bar{x}}_3[k] + e_{\bar{a}}[k]\,\bar{f}_d[k]$$
$$\delta\bar{x}[k+1] = \delta\bar{x}[k] - (1-\lambda_c)\,\delta\hat{\bar{x}}_3[k] - e_{\bar{a}}[k]\,\bar{f}_d[k] - \bar{a}[k]\,\bar{f}_T[k]$$
$$\delta\bar{x}[k+1] = \lambda_c\,\delta\bar{x}[k] + (1-\lambda_c)\,e_{\delta\bar{x}_3}[k] - e_{\bar{a}}[k]\,\bar{f}_d[k] - \bar{a}[k]\,\bar{f}_T[k]$$

3 True state of the process

State

$$\delta\bar{x}_1[k] = \delta\bar{x}[k-2], \qquad \delta\bar{x}_2[k] = \delta\bar{x}[k-1], \qquad \delta\bar{x}_3[k] = \delta\bar{x}[k] \qquad (d=2)$$

Change of the motion gain

$$\bar{a}[k+1] \approx \bar{a}[k] + \bar{a}'[k]\,\big(\bar{x}[k+1] - \bar{x}[k]\big)$$
$$\bar{x}[k+1] - \bar{x}[k] = \bar{a}[k]\,\bar{f}_d[k] + \bar{a}[k]\,\bar{f}_T[k]$$
$$\bar{x}[k+1] - \bar{x}[k] = \hat{M}[k] + e_{\bar{a}}[k]\,\bar{f}_d[k] + \bar{a}[k]\,\bar{f}_T[k]$$

True state

$$\delta\bar{x}_1[k+1] = \delta\bar{x}_2[k]$$
$$\delta\bar{x}_2[k+1] = \delta\bar{x}_3[k]$$
$$\delta\bar{x}_3[k+1] = \lambda_c\,\delta\bar{x}_3[k] + (1-\lambda_c)\,e_{\delta\bar{x}_3}[k] - \bar{f}_d[k]\,e_{\bar{a}}[k] - \bar{a}[k]\,\bar{f}_T[k]$$
$$\bar{a}[k+1] = \bar{a}[k] + \bar{a}'[k]\,\hat{M}[k] + \bar{a}'[k]\,\bar{f}_d[k]\,e_{\bar{a}}[k] + \bar{a}'[k]\,\bar{a}[k]\,\bar{f}_T[k]$$
$$\bar{a}'[k+1] = \bar{a}'[k] + w_{\bar{a}'}[k]$$

4 Estimator

$$e_{x_1}[k] = \delta\bar{x}_m[k] - \delta\hat{\bar{x}}_1[k]$$
$$e_{a_m}[k] = \bar{a}_m[k] - \hat{\bar{a}}[k]$$

$$\delta\hat{\bar{x}}_1[k+1] = \delta\hat{\bar{x}}_2[k] + \big\{\ell_{11}[k]\,e_{x_1}[k] + \ell_{12}[k]\,e_{a_m}[k]\big\}$$
$$\delta\hat{\bar{x}}_2[k+1] = \delta\hat{\bar{x}}_3[k] + \big\{\ell_{21}[k]\,e_{x_1}[k] + \ell_{22}[k]\,e_{a_m}[k]\big\}$$
$$\delta\hat{\bar{x}}_3[k+1] = \lambda_c\,\delta\hat{\bar{x}}_3[k] + \big\{\ell_{31}[k]\,e_{x_1}[k] + \ell_{32}[k]\,e_{a_m}[k]\big\}$$
$$\hat{\bar{a}}[k+1] = \hat{\bar{a}}[k] + \hat{\bar{a}}'[k]\,\hat{M}[k] + \big\{\ell_{41}[k]\,e_{x_1}[k] + \ell_{42}[k]\,e_{a_m}[k]\big\}$$
$$\hat{\bar{a}}'[k+1] = \hat{\bar{a}}'[k] + \big\{\ell_{51}[k]\,e_{x_1}[k] + \ell_{52}[k]\,e_{a_m}[k]\big\}$$

5 Dynamics of estimation errors

$$\mathbf{e}[k] = \begin{bmatrix} e_{\delta\bar{x}_1}[k] \\ e_{\delta\bar{x}_2}[k] \\ e_{\delta\bar{x}_3}[k] \\ e_{\bar{a}}[k] \\ e_{\bar{a}'}[k] \end{bmatrix} = \begin{bmatrix} \delta\bar{x}_1[k] - \delta\hat{\bar{x}}_1[k] \\ \delta\bar{x}_2[k] - \delta\hat{\bar{x}}_2[k] \\ \delta\bar{x}_3[k] - \delta\hat{\bar{x}}_3[k] \\ \bar{a}[k] - \hat{\bar{a}}[k] \\ \bar{a}'[k] - \hat{\bar{a}}'[k] \end{bmatrix}$$

Rows

$$e_{\delta\bar{x}_1}[k+1] = e_{\delta\bar{x}_2}[k] - \big\{\ell_{11}[k]\, e_{x_1}[k] + \ell_{12}[k]\, e_{a_m}[k]\big\}$$
$$e_{\delta\bar{x}_2}[k+1] = e_{\delta\bar{x}_3}[k] - \big\{\ell_{21}[k]\, e_{x_1}[k] + \ell_{22}[k]\, e_{a_m}[k]\big\}$$
$$e_{\delta\bar{x}_3}[k+1] = \lambda_c\, \delta\bar{x}_3[k] + (1-\lambda_c)\, e_{\delta\bar{x}_3}[k] - \lambda_c\, \delta\hat{\bar{x}}_3[k] - \bar{f}_d[k]\, e_{\bar{a}}[k] - \bar{a}[k]\, \bar{f}_T[k] - \big\{\ell_{31}[k]\, e_{x_1}[k] + \ell_{32}[k]\, e_{a_m}[k]\big\}$$
$$e_{\delta\bar{x}_3}[k+1] = e_{\delta\bar{x}_3}[k] - \bar{f}_d[k]\, e_{\bar{a}}[k] - \bar{a}[k]\, \bar{f}_T[k] - \big\{\ell_{31}[k]\, e_{x_1}[k] + \ell_{32}[k]\, e_{a_m}[k]\big\}$$
$$e_{\bar{a}}[k+1] = e_{\bar{a}}[k] + \hat{M}[k]\, e_{\bar{a}'}[k] + \bar{a}'[k]\, \bar{f}_d[k]\, e_{\bar{a}}[k] + \bar{a}'[k]\, \bar{a}[k]\, \bar{f}_T[k] - \big\{\ell_{41}[k]\, e_{x_1}[k] + \ell_{42}[k]\, e_{a_m}[k]\big\}$$
$$e_{\bar{a}}[k+1] \approx \big(1 + \hat{\bar{a}}'[k]\, \bar{f}_d[k]\big)\, e_{\bar{a}}[k] + \hat{M}[k]\, e_{\bar{a}'}[k] + \bar{a}'[k]\, \bar{a}[k]\, \bar{f}_T[k] - \big\{\ell_{41}[k]\, e_{x_1}[k] + \ell_{42}[k]\, e_{a_m}[k]\big\}$$
$$e_{\bar{a}'}[k+1] = e_{\bar{a}'}[k] + w_{\bar{a}'}[k] - \big\{\ell_{51}[k]\, e_{x_1}[k] + \ell_{52}[k]\, e_{a_m}[k]\big\}$$

Matrix form

$$\mathbf{e}[k+1] = \mathbf{F}_e[k]\, \mathbf{e}[k] - \mathbf{L}[k]\, \mathbf{H}\, \mathbf{e}[k] + \mathbf{q}[k]$$

$$\mathbf{F}_e[k] = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & -\bar{f}_d[k] & 0 \\ 0 & 0 & 0 & 1 + \hat{\bar{a}}'[k]\, \bar{f}_d[k] & \hat{M}[k] \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}, \qquad \mathbf{q}[k] = \begin{bmatrix} 0 \\ 0 \\ -\bar{a}[k]\, \bar{f}_T[k] \\ \bar{a}'[k]\, \bar{a}[k]\, \bar{f}_T[k] \\ w_{\bar{a}'}[k] \end{bmatrix}$$

Estimator feedback matrix and error covariance

$$\mathbf{L}[k] = \mathbf{P}^f[k]\, \mathbf{H}^T \big(\mathbf{H}\, \mathbf{P}^f[k]\, \mathbf{H}^T + \mathbf{R}[k]\big)^{-1}$$
$$\mathbf{P}[k] = \big(\mathbf{I} - \mathbf{L}[k]\, \mathbf{H}\big)\, \mathbf{P}^f[k]$$
$$\mathbf{P}^f[k+1] = \mathbf{F}_e[k]\, \mathbf{P}[k]\, \mathbf{F}_e^T[k] + \mathbf{Q}[k]$$

6 Process outputs and output matrix

Deterministic and random components of the measured tracking error

$$\overline{\delta\bar{x}_m}[k] = a_{\mathrm{pd}} \cdot \delta\bar{x}_m[k] + (1 - a_{\mathrm{pd}})\, \overline{\delta\bar{x}_m}[k-1]$$
$$\delta\bar{x}_r[k] = \delta\bar{x}_m[k] - \overline{\delta\bar{x}_m}[k]$$

Measured input gain

$$\bar{a}_m[k] = \frac{\delta\bar{x}_r^2[k] - C_n\,\sigma_n^2}{C_T\,\kappa_T}$$

$$\kappa_T = \frac{4k_BT\,a_o}{R}$$

$$C_T = (1-a_{\mathrm{pd}})^2\left[\frac{2\,(1-a_{\mathrm{pd}})(1-\lambda_c)}{1-(1-a_{\mathrm{pd}})\lambda_c} + \frac{2}{(2-a_{\mathrm{pd}})(1+\lambda_c)\big(1-(1-a_{\mathrm{pd}})\lambda_c\big)}\right]$$

$$C_n = \frac{2\,(1-a_{\mathrm{pd}})^2}{2-a_{\mathrm{pd}}}\left[1 + \frac{(1-a_{\mathrm{pd}})^2\,a_{\mathrm{pd}}\,(1-\lambda_c)}{1-(1-a_{\mathrm{pd}})\lambda_c} + \frac{(1-\lambda_c)^2}{(1+\lambda_c)\big(1-(1-a_{\mathrm{pd}})\lambda_c\big)}\right]$$

Measurement and output matrix

$$\mathbf{y}[k] = \begin{bmatrix}\delta\bar{x}_m[k]\\ \bar{a}_m[k]\end{bmatrix}, \qquad \hat{\mathbf{y}}[k] = \begin{bmatrix}\delta\hat{\bar{x}}_1[k]\\ \hat{\bar{a}}[k]\end{bmatrix}, \qquad \mathbf{H} = \begin{bmatrix}1&0&0&0&0\\0&0&0&1&0\end{bmatrix}$$

7 Process noise and measurement noise

Process noise

$$\mathbf{Q}[k] = \begin{bmatrix}0&0&0&0&0\\0&0&0&0&0\\0&0&q_{33}[k]&q_{34}[k]&0\\0&0&q_{43}[k]&q_{44}[k]&0\\0&0&0&0&q_{55}[k]\end{bmatrix}$$

$$q_{33}[k] = \kappa_T\,\hat{\bar{a}}[k]$$

$$q_{34}[k] = -\hat{\bar{a}}'[k]\,q_{33}[k]$$

$$q_{43}[k] = -\hat{\bar{a}}'[k]\,q_{33}[k]$$

$$q_{44}[k] = \hat{\bar{a}}'^2[k]\,q_{33}[k]$$

$$q_{55}[k] = P_{\mathrm{fix}}^2\,I[k]$$

$$I[k] = \frac{1}{p_{55}[k]} - \frac{1}{p_{55}^f[k]}$$

$$\sqrt{P_{\mathrm{fix}}} = 0.4\,\big(1.1/\bar{x}_d[k]\big)^2$$

Measurement noise

$$\mathbf{R}[k] = \mathrm{diag}\big(r_1,\ r_2[k]\big)$$

$$r_1 = \sigma_n^2$$

$$r_2[k] = \mathrm{IF}\cdot\sigma_{y_2}^2[k]$$

$$\sigma_{y_2}^2[k] = 2\,\big(\hat{\bar{a}}[k] + \xi\big)^2$$

$$\xi = \frac{C_n}{C_T}\,\frac{\sigma_n^2}{\kappa_T}$$

$$\mathrm{IF} = 1 + 2\sum_{\tau\geq 1}\rho_{y_2}(\tau)$$

8 Past-input law and the law of TIE 2025

Past-input law

$$\hat{\bar{x}}[k] = \bar{x}_m[k] + \sum_{i=1}^{d} \hat{\bar{a}}[k-i]\, \bar{f}_d[k-i]$$

$$\bar{f}_d[k] = \hat{\bar{a}}^{-1}[k] \left\{ \Delta \bar{x}_d[k] + (1-\lambda_c) \left(\bar{x}_d[k] - \hat{\bar{x}}[k] \right) \right\}$$

Law of TIE 2025

$$\bar{f}_d[k] = \hat{\bar{a}}^{-1}[k] \left\{ \Delta \bar{x}_d[k] + (1-\lambda_c)\, \delta \hat{\bar{x}}_3[k] \right\}$$